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PY211 Lab 4 — Projectile Motion

Objective

Determine the initial velocity v_0 of a steel ball launched horizontally from a spring gun and study the dependence of the projectile range R on the launch angle θ.

Equipment

- Spring gun with adjustable angle
- Steel ball
- Bar level
- Meter sticks
- Masking tape
- Paper (for marking landing positions)

Setup

The spring gun was clamped to the edge of a table at a measured height h above the floor. The angle of the gun could be adjusted using the built-in protractor, which reads the angle from the vertical. A sheet of paper was taped to the floor in the expected landing zone to record where the ball struck. The landing position was marked after each shot and its distance from the point directly below the launch point was measured with a meter stick to obtain the range R.

Process

1. Set the spring gun to $\theta_{\text{in}} = 0°$ from vertical (horizontal launch) and measure the height h from the center of the ball to the floor.
2. Fire the ball 5 times, recording the range R for each shot.
3. Compute the mean range $\overline{R}$, its standard deviation s_R, and the standard error of the mean $\delta\overline{R} = s_R/\sqrt{n}$.
4. Use $\overline{R}$ and h to calculate the initial velocity v_0.
5. Adjust the spring gun to angles of 10°, 15°, 20°, 25°, 30°, and 35° from vertical. At each angle, measure h and fire the ball 3 times, recording the range each time.
6. For each angle, compute the measured mean range and its uncertainty, the calculated range R_{calc} from the theoretical equation, and the propagated uncertainty δR_{calc}.
7. Compare measured and calculated ranges using residuals and z-scores.

Theory

For a horizontal launch ($\theta = 0°$ from horizontal), the initial velocity is:

$$v_0 = \overline{R}\sqrt{\frac{g}{2h}}$$

where $g = 9.81$ m/s^2 is the gravitational acceleration and h is the launch height.

The uncertainty on v_0 is obtained by error propagation:

$$\delta v_0 = v_0\sqrt{\left(\frac{\delta\overline{R}}{\overline{R}}\right)^2 + \frac{1}{4}\left(\frac{\delta h}{h}\right)^2}$$

For a launch at angle θ from the horizontal (where $\theta = 90° - \theta_{\text{in}}$ and θ_{in} is the angle read from the apparatus, measured from vertical), the theoretical range is:

$$R_{\text{calc}} = \frac{v_0^2\cos\theta}{g}\left(\sin\theta + \sqrt{\sin^2\theta + \frac{2gh}{v_0^2}}\right)$$

The uncertainty on R_{calc} is propagated from the uncertainties on v_0, h, and θ:

$$\delta R_{\text{calc}} = \sqrt{\left(\frac{\partial R}{\partial v_0}\,\delta v_0\right)^2 + \left(\frac{\partial R}{\partial h}\,\delta h\right)^2 + \left(\frac{\partial R}{\partial \theta}\,\delta\theta\right)^2}$$

where the partial derivatives were evaluated numerically.

Assumptions

- The gravitational acceleration is $g = 9.81$ m/s^2.
- Air resistance is negligible.
- The initial velocity v_0 is the same for every shot regardless of the launch angle (the spring compression is constant).
- The protractor on the apparatus reads the angle from the vertical accurately to within $\pm 0.5°$.
- The ball behaves as a point mass launched from a single well-defined point.

Sources of Experimental Uncertainty & Steps to Minimize

- Variation in spring release from shot to shot $\rightarrow$ practiced firing the gun consistently before recording data; took multiple shots at each angle.

- Difficulty reading the exact landing position → placed paper on the floor and marked each impact point; measured from a consistent reference line directly below the launch point.
- Uncertainty in height measurement ($\delta h = 0.005$ m) → measured h carefully with a meter stick at each angle setting, since tilting the gun changes the height of the ball's center.
- Uncertainty in angle reading ($\delta\theta = 0.5°$) → used the bar level to verify the table was level; read the protractor scale as precisely as possible.

Results of Experiment

Table 1: Calibration at $\theta_{\text{in}} = 0°$ (horizontal launch)

θ (°)	h (m)	n	$\overline{R}_{\text{meas}}$ (m)	$\delta\overline{R}$ (m)	v_0 (m/s)	δv_0 (m/s)
0	1.015	5	1.816	0.0103	3.992	0.0247

Raw range measurements at $\theta = 0°$: 1.84, 1.83, 1.81, 1.78, 1.82 m.

Table 2: Range vs. angle

θ_{in} (°)	θ_{h} (°)	h (m)	$\overline{R}_{\text{meas}}$ (m)	$\delta\overline{R}$ (m)	R_{calc} (m)	δR_{calc} (m)	z
10	80	1.035	2.010	0.0029	0.700	0.0347	37.6
15	75	1.040	2.068	0.0044	1.032	0.0337	30.5
20	70	1.050	2.190	0.0076	1.342	0.0325	25.4
25	65	1.070	2.288	0.0093	1.626	0.0310	20.5
30	60	1.080	2.330	0.0058	1.875	0.0293	15.2
35	55	1.090	2.343	0.0067	2.085	0.0277	9.07

Here θ_{in} is the angle from vertical read on the apparatus, $\theta_{\text{h}} = 90° - \theta_{\text{in}}$ is the angle from horizontal used in the range equation, and $z = (\overline{R}_{\text{meas}} - R_{\text{calc}})/\sqrt{(\delta\overline{R})^2 + (\delta R_{\text{calc}})^2}$ is the z-score comparing the two.

Graphical Analysis

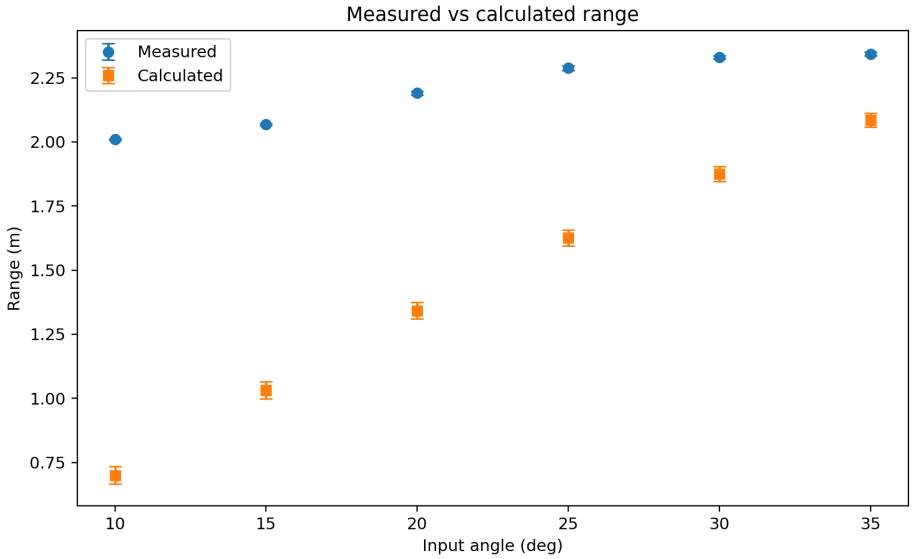

Figure 1: Measured and calculated range as a function of the input angle (from vertical). Error bars represent $\delta\overline{R}$ and δR_{calc} respectively.

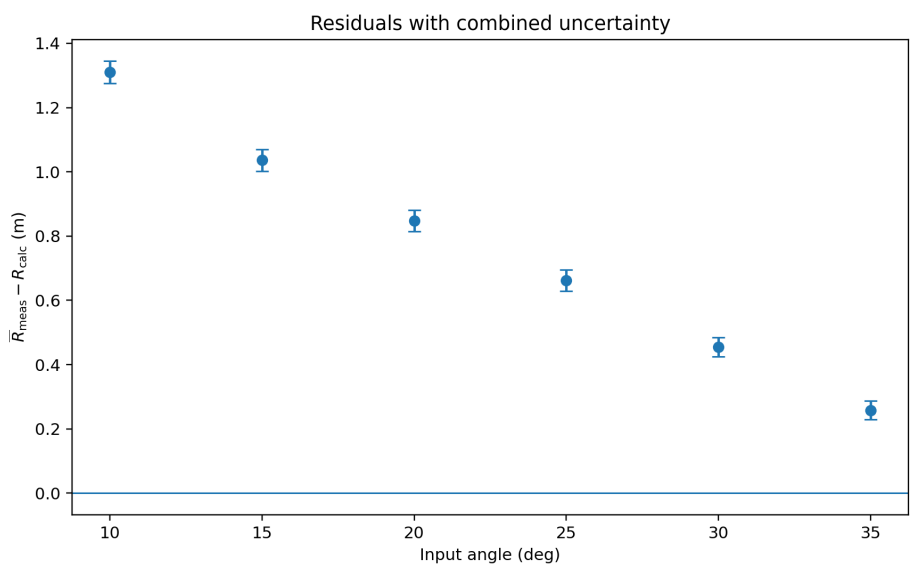

Figure 2: Residuals ($\overline{R}_{\text{meas}} - R_{\text{calc}}$) with combined uncertainty at each angle.

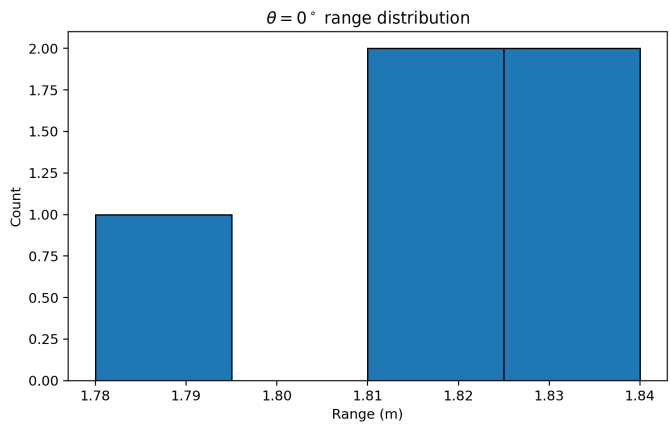

Figure 3: Distribution of the five range measurements at $\theta = 0°$.

Calculations

Detailed, step-by-step calculations with full substitutions and uncertainty propagation for every trial are provided in the attached document `projectile_lab_show_work.pdf`.

As a representative example, the calibration velocity:

$$v_0 = (1.816)\sqrt{\frac{9.81}{2(1.015)}} = 3.992 \text{ m/s}$$

$$\delta v_0 = (3.992)\sqrt{\left(\frac{0.0103}{1.816}\right)^2 + \frac{1}{4}\left(\frac{0.005}{1.015}\right)^2} = 0.0247 \text{ m/s}$$

Conclusion

The initial velocity of the steel ball was found to be $v_0 = (3.992 \pm 0.025)$ m/s. At all six angles (10°–35° from vertical), the measured ranges exceeded the calculated values, yielding z-scores between 9 and 38—well above the consistency threshold of 2. The discrepancy shrinks as the gun approaches horizontal, suggesting a systematic effect tied to the tilt of the barrel, such as a change in effective launch height h or reference point when the gun is angled. These effects are not reflected in our statistical uncertainties. The qualitative trend is nonetheless correct: range increases monotonically as the angle approaches horizontal, consistent with theory. More careful measurement of h and the range reference at each angle setting would reduce the systematic offset.